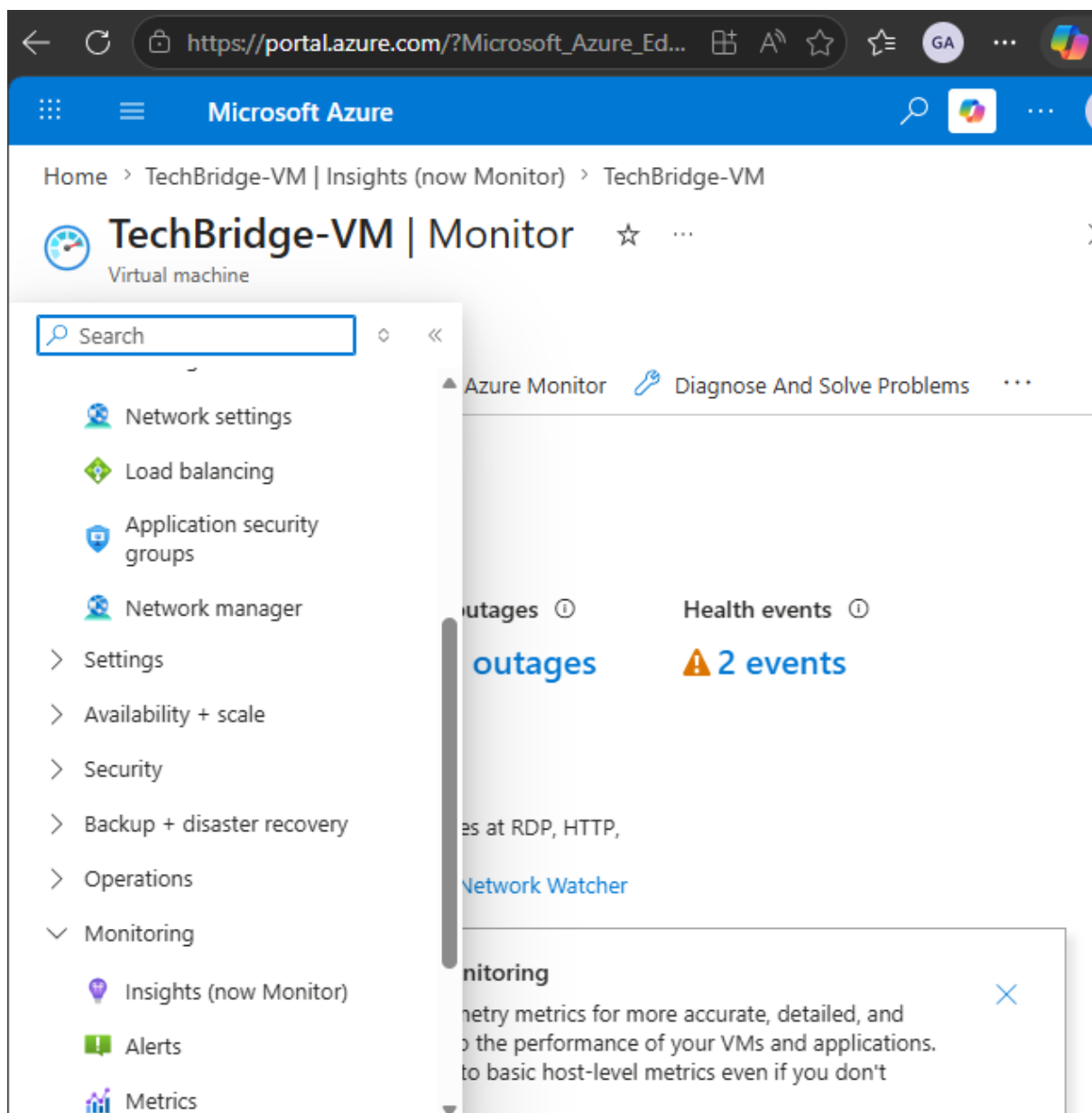


Week 7

Task 1

I enabled VM Insights on my TechBridge-VM, which automatically configures a Log Analytics Workspace named LogAnalytics-20240435 in the same region as my VM. The collected data includes CPU metrics, memory usage, disk I/O, and network traffic, all of which are stored in the Log Analytics Workspace with a default retention period of 30 days for metrics and 90 days for logs (free tier). VM Insights provides a pre-built dashboard with performance charts and dependency maps, making it easier to visualize resource health without building custom queries.



Diagnostic setting ...



Save Discard Delete Feedback

want to collect from a resource, and one or more destinations that you would stream them to. Normal usage charges for the destination will occur. [Learn more about the different log categories and contents of those logs](#)

Diagnostic setting name *

NSG-Diagnostics-20240435

Logs

Category groups ⓘ

☐ allLogs

Categories

☒ Network Security Group Event

☒ Network Security Group Rule Counter

Destination details

☒ Send to Log Analytics workspace

Subscription

Azure for Students

Log Analytics workspace

No workspaces in this subscription.

☐ Archive to a storage account

☐ Stream to an event hub

☐ Send to partner solution

Diagnostic setting ... ✕

Save
 Discard
 Delete
 Feedback

A diagnostic setting specifies a list of categories of platform logs and/or metrics that you want to collect from a resource, and one or more destinations that you would stream them to. Normal usage charges for the destination will occur. [Learn more about the different log categories and contents of those logs](#)
JSON View

Diagnostic setting name *

VNet-Diagnostics-20240435 ✓

Logs

Category groups ?

☐ allLogs

Categories

☒ VM protection alerts

Metrics

☐ AllMetrics

Destination details

☒ Send to Log Analytics workspace

Subscription

Azure for Students ▼

Log Analytics workspace

No workspaces in this subscription. ▼

☐ Archive to a storage account

☐ Stream to an event hub

Task 3

Resources Monitored

Resource Name	Resource Type	Monitoring Method	Data Collected
TechBridge-VM	Virtual Machine	VM Insights + Diagnostics Settings	CPU %, Memory, Disk I/O, Network In/Out, OS logs

Resource Name	Resource Type	Monitoring Method	Data Collected
NSG-20240435-Week5	Network Security Group	Diagnostic Settings (Event category)	All NSG rule matches (allowed/denied traffic)
TechBridge-VNet	Virtual Network	Attempted (limited on free tier)	No data - documented limitation

Unexpected CPU spikes could indicate crypto mining malware, scheduled backup jobs running at wrong times, or memory leaks causing garbage collection. Unexpected network outbound spikes could indicate data exfiltration attempts. Unexpected disk writes could indicate ransomware encryption activity or excessive logging.

Task 4

Monitoring data directly helps us save money by finding resources that are not being used enough and resources that are being used much. For example if we look at the CPU metrics and we see that the average usage is below 5 percent over 30 days then we can justify downsizing a machine from B2s to B1s or turning it off during non-business hours using auto-shutdown. Monitoring data also helps us with network metrics. If we see patterns of sending data out then we can find ways to save money. We can move data that we access a lot to storage options like Cool or Archive. We can also use Azure CDN to reduce the costs of sending data out. Monitoring data helps us with disk metrics too. If we see that we are not doing input and output operations on our premium solid state drives then we can move to standard hard disk drives for the work that we do when we are not making anything for people to use. We can also look at the logs from our network security groups and firewalls. This helps us find traffic patterns that we do not expect.